

Evidence 1c-a

Deliverables Checklist

- [] **Terraform Plan:** Console output showing all resources to be created.
 - terraform plan output
 - terraform apply evidence (outputs)
 - [] **Private Connectivity (Bonus A):** EC2 has no Public IP; VPC Endpoints used for SSM and Secrets Manager.
 - CLI verification commands (from Lab 1b)
 - Incident runbook execution notes (alarm fired + recovered)
-

Terraform Plan: Console Output

```
ch887@LAPTOP-GCPD1732 MINGW64 ~/Documents/Class7_Armageddon/Armageddon v0.41 (Van)/DB - Prototype/LAB1/c/secrets (Daniel-Bryce-Armageddon-Branch)
$ terraform plan
aws_secretsmanager_secret.rds_secret: Refreshing state... [id=arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab-1c/rds/mysql-8uwlwN]
aws_secretsmanager_secret_version.rds_secret_version: Refreshing state... [id=arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab-1c/rds/mysql-8uwlwN|terraform-20260203190212758800000002]
```

terraform plan -var-file="lab-1c.tfvars"

module.iam.aws_iam_policy.cloudwatch_logs: Refreshing state...

[id=arn:aws:iam::619076214091:policy/armageddon-lab-1c-cloudwatch-logs-policy]

module.vpc.aws_vpc.main: Refreshing state... [id=vpc-0d0013e86f325ffc4]

module.config_store.aws_secretsmanager_secret.db_credentials: Refreshing state...

[id=arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab/rds/mysql2-c9a0CQ]

module.cloudwatch.aws_sns_topic.db_incidents: Refreshing state...

[id=arn:aws:sns:ap-northeast-1:619076214091:lab-db-incidents]

module.iam.aws_iam_policy.ssm_read: Refreshing state...

[id=arn:aws:iam::619076214091:policy/armageddon-lab-1c-ssm-read-policy]

module.cloudwatch.aws_cloudwatch_log_group.app_logs: Refreshing state...

[id=/aws/ec2/lab-rds-app]

data.aws_secretsmanager_secret.rds: Reading...

module.iam.aws_iam_policy.ec2_secrets_policy: Refreshing state...

[id=arn:aws:iam::619076214091:policy/armageddon-lab-1c-EC2ReadRDSecret]

module.iam.aws_iam_role.ec2_secrets_role: Refreshing state...

[id=armageddon-lab-1c-ec2-secrets-role]

module.iam.aws_iam_instance_profile.ec2_profile: Refreshing state...

[id=armageddon-lab-1c-ec2-secrets-profile]

module.iam.aws_iam_role_policy_attachment.cw_agent: Refreshing state...
[id=armageddon-lab-1c-ec2-secrets-role/arn:aws:iam::aws:policy/CloudWatchLogsFullAccess]

module.iam.aws_iam_role_policy_attachment.ssm_read: Refreshing state...
[id=armageddon-lab-1c-ec2-secrets-role/arn:aws:iam::619076214091:policy/armageddon-lab-1c-ssm-read-policy]

module.iam.aws_iam_role_policy.ec2_policy: Refreshing state...
[id=armageddon-lab-1c-ec2-secrets-role:ec2_policy]

module.iam.aws_iam_role_policy_attachment.secrets_attach: Refreshing state...
[id=armageddon-lab-1c-ec2-secrets-role/arn:aws:iam::619076214091:policy/armageddon-lab-1c-EC2ReadRDSecret]

data.aws_secretsmanager_secret.rds: Read complete after 2s
[id=arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab-1c/rds/mysql-8uwkwN]

module.iam.aws_iam_role_policy_attachment.cloudwatch_logs: Refreshing state...
[id=armageddon-lab-1c-ec2-secrets-role/arn:aws:iam::619076214091:policy/armageddon-lab-1c-cloudwatch-logs-policy]

data.aws_secretsmanager_secret_version.rds: Reading...

data.aws_secretsmanager_secret_version.rds: Read complete after 0s
[id=arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab-1c/rds/mysql-8uwkwN|AWSCURRENT]

module.config_store.aws_ssm_parameter.db_endpoint: Refreshing state...
[id=/lab/db/endpoint]

module.config_store.aws_ssm_parameter.db_port: Refreshing state... [id=/lab/db/port]

module.config_store.aws_ssm_parameter.db_name: Refreshing state... [id=/lab/db/name]

module.cloudwatch.aws_sns_topic_subscription.email: Refreshing state...
[id=arn:aws:sns:ap-northeast-1:619076214091:lab-db-incidents:230b5fca-f2be-4c38-a959-a5ea5467f993]

module.config_store.aws_secretsmanager_secret_version.db_credentials_version: Refreshing state...
[id=arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab/rds/mysql2-c9a0CQ|terraform-20260203193722935600000002]

module.cloudwatch.aws_cloudwatch_log_metric_filter.db_errors: Refreshing state...
[id=DBConnectionErrors]

module.cloudwatch.aws_cloudwatch_metric_alarm.db_failure: Refreshing state...
[id=lab-db-connection-failure]

module.vpc.aws_route_table.private: Refreshing state... [id=rtb-0a1b1ca691ece2c01]

module.vpc.aws_subnet.public_a: Refreshing state... [id=subnet-071fa792b93a9baa6]

module.vpc.aws_subnet.private_a: Refreshing state... [id=subnet-06008505223344b58]

module.vpc.aws_internet_gateway.main: Refreshing state... [id=igw-0cbb12be7784dcd6]

module.vpc.aws_subnet.private_b: Refreshing state... [id=subnet-09dcfa3f69e5c8a37]

module.security.aws_security_group.rds_sg: Refreshing state... [id=sg-007339b479c471df1]

module.security.aws_security_group.ec2_sg: Refreshing state... [id=sg-0737f17f6bada27e6]

module.security.aws_vpc_security_group_ingress_rule.ec2_http: Refreshing state...
[id=sgr-02252912069405c99]

module.security.aws_vpc_security_group_egress_rule.ec2_all_outbound: Refreshing state...
[id=sgr-00624285719a3ff68]

module.vpc.aws_eip.nat: Refreshing state... [id=eipalloc-0071c3b55f3bb2686]

module.vpc.aws_route_table.public: Refreshing state... [id=rtb-010630fb323c0cc19]

module.security.aws_vpc_security_group_egress_rule.rds_all_outbound: Refreshing state...
[id=sgr-02e8ea5de8829f75f]

module.security.aws_vpc_security_group_ingress_rule.rds-http_ipv4: Refreshing state...
[id=sgr-06f5c368d71bc360b]

module.vpc.aws_route_table_association.private_a: Refreshing state...
[id=rtbassoc-0350c8c5ebb28c4db]

module.vpc.aws_db_subnet_group.this: Refreshing state...
[id=armageddon-lab-1c-db-subnet-group]

module.ec2.aws_instance.ec2: Refreshing state... [id=i-09b1034b617e7c03c]

module.vpc.aws_route_table_association.public_a: Refreshing state...
[id=rtbassoc-0aa9fce11d9e7fcee]

module.vpc.aws_nat_gateway.main: Refreshing state... [id=nat-0cb9bbdaacc9ecceb]

module.rds.aws_db_instance.mysql: Refreshing state...
[id=db-4ANRC373NZKGCC4JUHYXVZ2FU4]

module.rds.aws_ssm_parameter.db_config["endpoint"]: Refreshing state...
[id=/lab-1c/db/endpoint]

module.rds.aws_ssm_parameter.db_config["name"]: Refreshing state... [id=/lab-1c/db/name]

module.rds.aws_ssm_parameter.db_config["port"]: Refreshing state... [id=/lab-1c/db/port]

Private Check: Private EC2

```
~ $ aws ec2 describe-instances --instance-ids <i-09b1034b617e7c03c> --query "Reservations[].Instances[].PublicIpAddress"
-bash: i-09b1034b617e7c03c: No such file or directory
~ $ aws ec2 describe-instances --instance-ids i-09b1034b617e7c03c --query "Reservations[].Instances[].PublicIpAddress"
[]
~ $ aws ec2 describe-instances --instance-ids i-09b1034b617e7c03c --query "Reservations[].Instances[].State.Name"
[
  "running"
]
```

Endpoints: VPC Endpoints

```
ap-northeast-1 +
~ $ aws ec2 describe-vpc-endpoints --region ap-northeast-1 --query "VpcEndpoints[].ServiceName"
[
  "com.amazonaws.ap-northeast-1.s3",
  "com.amazonaws.ap-northeast-1.ec2messages",
  "com.amazonaws.ap-northeast-1.ec2",
  "com.amazonaws.ap-northeast-1.secretsmanager",
  "com.amazonaws.ap-northeast-1.logs",
  "com.amazonaws.ap-northeast-1.ssmmessages",
  "com.amazonaws.ap-northeast-1.ssm"
]
```

Session Manager: EC2 Instance appears private

```
~ $ aws ssm describe-instance-information --query "InstanceInformationList[].InstanceId"
[
  "i-04e4d1c825fb35b7e"
]
```

Evidence 1c-b

- [] **Load Balanced (Bonus B):** Application Load Balancer (ALB) with WAF and TLS certificate (ACM).
- [] **Persistence:** S3 Bucket for ALB access logs with appropriate bucket policy.
- [] **ALB Active State:** Prove the load balancer is provisioned and running.
- [] **HTTPS Listener:** Prove a listener exists on port 443 with the ACM certificate attached.
- [] **Target Health:** Prove the private EC2 instances are reporting as **healthy** in the Target Group.
- [] **WAF Association:** Prove the Web ACL is successfully protecting the ALB.
- [] **Alarm & Dashboard:** Show that the ALB 5xx alarm and the CloudWatch dashboard exist.

- [] **ALB Active State:** Prove the load balancer is provisioned and running

```
{
  "LoadBalancers": [
    {
      "LoadBalancerArn": "arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:loadbalancer/app/armageddon-lab-1c-alb-public/92a9dac0b1302ec0",
      "DNSName": "armageddon-lab-1c-alb-public-2011588908.ap-northeast-1.elb.amazonaws.com",
      "CanonicalHostedZoneId": "Z14GRHDQWA56QT",
      "CreatedTime": "2026-02-09T20:25:55.636000+00:00",
      "LoadBalancerName": "armageddon-lab-1c-alb-public",
      "Scheme": "internet-facing",
      "VpcId": "vpc-0024cd2acddec3381",
      "State": {
        "Code": "active"
      },
      "Type": "application",
      "AvailabilityZones": [
        {
          "ZoneName": "ap-northeast-1a",
          "SubnetId": "subnet-082d49ecd3a8968f2",
          "LoadBalancerAddresses": []
        },
        {
          "ZoneName": "ap-northeast-1c",
          "SubnetId": "subnet-0f6688472176a866f",
          "LoadBalancerAddresses": []
        }
      ]
    }
  ],
}
```

```
{
  "ListenerArn": "arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:listener/app/armageddon-lab-1c-alb-public/92a9dac0b1302ec0/934a274ecd7bd4c4",
  "LoadBalancerArn": "arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:loadbalancer/app/armageddon-lab-1c-alb-public/92a9dac0b1302ec0",
  "Port": 443,
  "Protocol": "HTTPS",
  "Certificates": [
    {
      "CertificateArn": "arn:aws:acm:ap-northeast-1:619076214091:certificate/15e3ed5-74e9-4a6d-974b-820ef04f0ca4"
    }
  ],
  "SslPolicy": "ELBSecurityPolicy-TLS13-1-2-2021-06",
  "DefaultActions": [
    {
      "Type": "forward",
      "TargetGroupArn": "arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:targetgroup/armageddon-lab-1c-alb-pr/4442da45e4415527",
      "Order": 1,
      "ForwardConfig": {
        "TargetGroups": [
          {
            "TargetGroupArn": "arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:targetgroup/armageddon-lab-1c-alb-pr/4442da45e4415527",
            "Weight": 1
          }
        ]
      }
    }
  ]
}
```

- [] **Persistence:** S3 Bucket for ALB access logs with appropriate bucket policy.
- [] **Target Health:** Prove the private EC2 instances are reporting as **healthy** in the Target Group.

```
- $ aws elbv2 describe-target-groups \
> --load-balancer-arn arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:loadbalancer/app/armageddon-lab-1c-alb-public/92a9dac0b1302ec0 \
> --query "TargetGroups[0].TargetGroupArn" --output text
arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:targetgroup/armageddon-lab-1c-alb-pr/4442da45e4415527
- $ aws elbv2 describe-target-health \
> --target-group-arn arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:targetgroup/armageddon-lab-1c-alb-pr/4442da45e4415527
{
  "TargetHealthDescriptions": [
    {
      "Target": {
        "Id": "i-055e111ae28f8db35",
        "Port": 80
      },
      "HealthCheckPort": "80",
      "TargetHealth": {
        "State": "unhealthy",
        "Reason": "Target.FailedHealthChecks",
        "Description": "Health checks failed"
      },
      "AdministrativeOverride": {
        "State": "no override",
        "Reason": "AdministrativeOverride.NoOverride",
        "Description": "No override is currently active on target"
      }
    }
  ]
}
- $
```

```
> --target-group-arn arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:targetgroup/armageddon-lab-1c-alb-pr/4442da45e4415527 \
> --query "TargetHealthDescriptions[].{InstanceID:Target.Id, Status:TargetHealth.State, Reason:TargetHealth.Reason}" \
> --output table
```

DescribeTargetHealth		
InstanceID	Reason	Status
i-055e111ae28f8db35	None	healthy

```
~ $
```

- [] **WAF Association:** Prove the Web ACL is successfully protecting the ALB.

```
aws wafv2 get-web-acl-for-resource \
```

```
> --resource-arn
arn:aws:elasticloadbalancing:ap-northeast-1:619076214091:loadbalancer/app/armageddon-lab-1c-alb-public/92a9dac0b1302ec0

{
  "WebACL": {
    "Name": "lab-1c-waf01",
    "Id": "32497572-7a34-415e-93cd-fdd24fceb188",
    "ARN":
"arn:aws:wafv2:ap-northeast-1:619076214091:regional/webacl/lab-1c-waf01/32497572-7a34-415e-93cd-fdd24fceb188",
    "DefaultAction": {
      "Allow": {}
    },
    "Description": "",
    "Rules": [
      {
        "Name": "AWSManagedRulesCommonRuleSet",
        "Priority": 1,
        "Statement": {
          "ManagedRuleGroupStatement": {
            "VendorName": "AWS",
            "Name": "AWSManagedRulesCommonRuleSet"
```

```

    }

  },

  "OverrideAction": {

    "None": {}

  },

  "VisibilityConfig": {

    "SampledRequestsEnabled": true,

    "CloudWatchMetricsEnabled": true,

    "MetricName": "lab-1c-waf-common"

  }

:...skipping...

{

  "WebACL": {

    "Name": "lab-1c-waf01",

    "Id": "32497572-7a34-415e-93cd-fdd24fceb188",

    "ARN":
"arn:aws:wafv2:ap-northeast-1:619076214091:regional/webacl/lab-1c-waf01/32497572-7a34-415e-93cd-fdd24fceb188",

    "DefaultAction": {

      "Allow": {}

    },

    "Description": "",

    "Rules": [

      {

        "Name": "AWSManagedRulesCommonRuleSet",

        "Priority": 1,

        "Statement": {

```

```
    "ManagedRuleGroupStatement": {
      "VendorName": "AWS",
      "Name": "AWSManagedRulesCommonRuleSet"
    }
  },
  "OverrideAction": {
    "None": {}
  },
  "VisibilityConfig": {
    "SampledRequestsEnabled": true,
    "CloudWatchMetricsEnabled": true,
    "MetricName": "lab-1c-waf-common"
  }
}
],
"VisibilityConfig": {
  "SampledRequestsEnabled": true,
  "CloudWatchMetricsEnabled": true,
  "MetricName": "lab-1c-waf01"
},
"Capacity": 700,
"ManagedByFirewallManager": false,
"LabelNamespace": "aws:waf:619076214091:webacl:lab-1c-waf01:",
"RetrofittedByFirewallManager": false,
"OnSourceDDoSProtectionConfig": {
  "ALBLowReputationMode": "ACTIVE_UNDER_DDOS"
```



```
}
}
}
```

- [] **Alarm & Dashboard:** Show that the ALB 5xx alarm and the CloudWatch dashboard exist.

```
"MetricAlarms": [
  {
    "AlarmName": "TestAlarmPostDBStop",
    "AlarmArn": "arn:aws:cloudwatch:ap-northeast-1:619076214091:alarm:TestAlarmPostDBStop",
    "AlarmDescription": "Alert when RDS has no connections (stopped/down)",
    "AlarmConfigurationUpdatedTimestamp": "2026-02-02T20:28:45.235000+00:00",
    "ActionsEnabled": true,
    "OKActions": [],
    "AlarmActions": [
      "arn:aws:sns:ap-northeast-1:619076214091:lab-db-incidents"
    ],
    "InsufficientDataActions": [],
    "StateValue": "ALARM",
    "StateReason": "Threshold Crossed: no datapoints were received for 1 period and 1 missing datapoint was treated as [Breaching].-",
    "StateReasonData": "{\n  \"version\": \"1.0\", \"queryDate\": \"2026-02-02T20:29:15.921+0000\", \"statistic\": \"Sum\", \"period\": 60, \"recentDatapoints\": [], \"threshold\": 1.0, \"evaluatedDatapoints\": [{\n    \"timestamp\": \"2026-02-02T20:28:00.000+0000\" } ] }",
    "StateUpdatedTimestamp": "2026-02-02T20:29:15.922000+00:00",
    "MetricName": "DatabaseConnections",
    "Namespace": "AWS/RDS",
    "Statistic": "Sum",
    "Dimensions": [
      {
        "Name": "DBInstanceIdentifier",
        "Value": "lab-mysql1"
      }
    ],
    "Period": 60,
    "EvaluationPeriods": 1,
    "DatapointsToAlarm": 1,
  }
]
```

```
CloudShell

ap-northeast-1 +

~ $ aws cloudwatch list-dashboards

{
  "DashboardEntries": [
    {
      "DashboardName": "lab-1c-dashboard01",
      "DashboardArn": "arn:aws:cloudwatch:619076214091:dashboard/lab-1c-dashboard01",
      "LastModified": "2026-02-09T20:28:43+00:00",
      "Size": 706
    }
  ]
}
```

Reflection Questions (Lab 1C)

- **A)** Why do we terminate TLS at the ALB instead of the EC2 instance?

Using AWS Certificate Manager (ACM) with the ALB allows for automated renewals.

- **B)** What is the security benefit of placing the EC2 instance in a private subnet while the ALB remains public?

Zero Direct Exposure: Even if an attacker finds your instance's IP, they cannot reach it via the public internet because there is no Route Table entry to an Internet Gateway (IGW).

- **C)** Why is DNS validation preferred over Email validation for ACM certificates in a Terraform workflow?

DNS validation covers all subdomains and facilitates easier renewals without tracking down whoever owns the "admin@company.com" inbox

- **D)** If your Target Health is **unhealthy**, list three common configuration errors you should check first.

Security Groups

Incorrect Health Check Path: Check the status of Python app via terminal scripts

Check App launched / User data correct: Did the **user_data** script fail to install dependencies (e.g., **pip install** timeout)